

Total No. of Questions :10]

SEAT No. :

P3434

[5670]-710

[Total No. of Pages : 2

B.E. (Computer Engg.)

INFORMATION AND CYBER SECURITY

(2015 Course) (Semester - II) (410251) (End Sem)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8, Q9 or Q10.
- 2) Neat diagrams must be drawn whenever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary.

Q1) a) Describe linear and differential cryptanalysis with suitable example [5]

b) Distinguish between Substitution and transposition ciphers [5]

OR

Q2) a) Explain Operational model of security in detail. [5]

b) What is Transposition Cipher? Use Transposition Cipher to encrypt the plain text "WE ARE THE BEST" use key "HEAVEN" [5]

Q3) a) Explain digital signature standard. [5]

b) Describe Elliptic Curve Cryptography [5]

OR

Q4) a) What is Block cipher? Explain counter mode of block cipher. [5]

b) Write note on application and limitation of stenography. [5]

Q5) a) What is Backdoors and Key Escrow in PGP? [9]

b) Describe IPsec protocol with its components and security services. [8]

OR

Q6) a) Explain OAKLEY key determination protocol. [9]

b) Explain Secure Electronic Transaction in detail. [8]

Q7) a) Describe briefly how IPsec works and enlist its applications. Distinguish between tunnel and transport mode of IPsec [8]

b) Explain in detail password management practices. [9]

OR

P.T.O.

- Q8)** a) What are various types of firewall? Discuss limitations of firewall. [9]
b) Explain need and challenges of intrusion detection system. Define signature based IDS. [8]
- Q9)** a) Discuss PII confidentiality safeguards. [8]
b) What is cyber stalking? How to identify and detect cyber stalking. [8]

OR

- Q10)** a) Explain personally identifiable information PII. Describe PII impact levels with examples [8]
b) Write note on Information Protection Law: Indian perspective [8]

